Project Proposal: Context-Aware Neural Speech Decoding with Skip-Diphone Auxiliary Supervision and Temporal Smoothness Regularization

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1 Motivation

Speech brain–computer interfaces (BCIs) decode attempted speech from intracortical neural activity into text. The standard pipeline maps multi-channel neural features to frame-level phoneme probabilities with a recurrent neural network (RNN/GRU) under CTC loss, followed by a language model that converts phoneme predictions into words [1]. Predicting isolated (mono)phonemes is suboptimal because speech production is strongly context-dependent (*coarticulation*). The recently proposed **DCoND** framework [3] addresses this by predicting *diphones* ($z_{t-1} \to z_t$) and marginalizing over the previous-phoneme dimension to recover phoneme probabilities, achieving state-of-the-art Phoneme Error Rate (PER) on the Brain-to-Text '24 Benchmark.

This project builds on DCoND and tests two extensions. **(i) A skip-diphone auxiliary head** predicting the wider context pair $z_{t-2} \to z_t$ in addition to the standard adjacent diphone. The hypothesis is that speech motor planning encodes context beyond the immediately preceding phoneme; e.g., for *cat* (`K AE T`), the standard diphones cover `K` $\to$ `AE` and `AE` $\to$ `T`, while `K` $\to$ `T` probes whether earlier articulatory context still shapes later neural patterns. The auxiliary head is used only for supervision and is discarded at inference. **(ii) A temporal smoothness loss** on the marginalized phoneme probabilities. Because intracortical recordings are noisy and non-stationary, frame-level outputs can be jittery and trigger spurious phoneme insertions under CTC decoding; a smoothness regularizer encourages stable trajectories without preventing legitimate transitions.

2 Planned Data

I will use the **Brain-to-Text '24 Benchmark** (Eval.AI Challenge #2099 [4]), based on the speech BCI dataset of Willett et al. [2] (Dryad DOI `10.5061/dryad.x69p8czpq`, *Version 4*). The dataset contains ~12,100 attempted-speech sentences from a single ALS participant (T12), recorded with intracortical microelectrode arrays in speech motor cortex; neural features are 20 ms-binned threshold-crossing rates and spike-band power. Specifically I will use: (a) `competitionData/` — official train/test/`competitionHoldOut` splits (24 recording sessions, April–August 2022), converted from `.mat` to `.pkl` via the `cffan/neural_seq_decoder` preprocessing notebook [5]; (b) `languageModel/` — the released 3-gram (and optionally 5-gram) phoneme-to-text LM for WER evaluation. Phoneme labels are obtained from sentence transcripts via the CMU Pronouncing Dictionary, following [1, 3]. My baseline codebase is the official PyTorch port `cffan/neural_seq_decoder` [5] by Fan (a co-author of [1]), which is the same codebase that DCoND [3] extends; the original TensorFlow release is `fwillett/speechBCI`. For debugging I will train on a small session subset, then scale to the full training set.

3 State-of-the-Art Approaches and Planned Selection

The NPTL baseline [1] is a bidirectional GRU mapping neural features to monophone probabilities under CTC; DCoND [3] replaces this target with a diphone target (C^2=1600 classes) and marginalizes back to phonemes. I follow DCoND's GRU encoder and CTC objective, and study the following ablation variants: **(A)** GRU + monophone CTC (NPTL baseline); **(B)** GRU + standard diphone with marginalization (DCoND); **(C)** B + temporal smoothness loss; **(D)** B + skip-diphone auxiliary head (predicting $z_{t-2} \to z_t$); **(E)** full model = B + skip-diphone aux. + smoothness. The training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}^{\text{CTC}}_{\text{phoneme}} + \alpha\,\mathcal{L}^{\text{CTC}}_{\text{std-diphone}} + \beta\,\mathcal{L}^{\text{CTC}}_{\text{skip-diphone}} + \lambda\,\mathcal{L}_{\text{smooth}}, \quad \mathcal{L}_{\text{smooth}} = \tfrac{1}{T-1}\sum_{t=2}^{T}\left\|p_t - p_{t-1}\right\|_2^2, \tag{1}$$

where p_t is the marginalized phoneme distribution at frame t. Following [3] I initialize α=0.6. I expect β and λ to be small so that auxiliary supervision and regularization *guide* — but do not dominate — the main CTC objective.

4 Evaluation Metrics

Primary metrics are **Phoneme Error Rate (PER)** and **Word Error Rate (WER)** after n-gram LM decoding, plus validation CTC loss for monitoring. I will report the five-variant ablation (A–E) above, sweep the smoothness weight $\lambda \in \{0,\ 10^{-3},\ 5{\cdot}10^{-3},\ 10^{-2}\}$, and run a small sweep over β. I hypothesize that moderate smoothness reduces frame-level jitter and improves PER/WER, while excessive smoothing erases phoneme boundaries; for the skip-diphone head, I will assess whether broader context helps under limited neural training data, or whether immediate diphones remain more reliable. Final-model results will also be submitted to the Eval.AI leaderboard [4] as an external sanity check.

Expected contribution. Beyond reproducing DCoND, this project tests whether (i) *broader phoneme context* via skip-diphone auxiliary supervision and (ii) *temporal output regularization* jointly improve neural speech decoding, yielding a context-aware, temporally regularized decoder for brain-to-text.

References **[1]** F. R. Willett *et al.*, A high-performance speech neuroprosthesis, *Nature* 620:1031–1036, 2023. **[2]** F. R. Willett *et al.*, Data for: A high-performance speech neuroprosthesis, *Dryad*, 2023, https://doi.org/10.5061/dryad.x69p8czpq. **[3]** J. Li, T. Le, C. Fan, M. Chen, E. Shlizerman, Brain-to-Text Decoding with Context-Aware Neural Representations and LLMs, *arXiv:2411.10657*, 2024. **[4]** Brain-to-Text Benchmark '24, Eval.AI Challenge #2099, https://eval.ai/web/challenges/challenge-page/2099/overview. **[5]** C. Fan *et al.*, Neural Sequence Decoder (PyTorch reference implementation), GitHub, https://github.com/cffan/neural_seq_decoder.